TALEX MAXIM

Staff AI Engineer · Security Researcher

PhD Researcher in Computer Science - University of Greenwich

Doctoral research: Chagu and agentic AI security · ATAS UK clearance.

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PROFESSIONAL PROFILE

Data Scientist and Staff AI Engineer with a background in statistics, machine learning, and production systems. Develop ML models, LLM applications, and AI agents, from defining the problem and testing an approach to deployment and ongoing evaluation.

Work spans anomaly detection, application security, data pipelines, and the infrastructure needed to run models reliably. Research how code, dependencies, permissions, and exposed features can create exploitable paths through a system.

Registered PhD Researcher at the University of Greenwich and creator of Chagu. Study security and accountability in agentic AI using anomaly detection, provenance graphs, and blockchain.

RESEARCH AND SELECTED PROJECTS

Chagu

Creator, AI Data Scientist and Researcher | August 2024-Present

- Created Chagu, a protocol and research project on security and accountability in distributed and agentic systems.
- Academic research forming the basis of my doctoral thesis at the University of Greenwich.
- Study anomaly detection, provenance graphs, and blockchain-based accountability.
- Build prototypes and repeatable experiments to test agent behavior, security assumptions, and failure cases.
- Share the work through open-source code, technical articles, and talks.

Security Beyond Chagu

Data science for application security and attack-path analysis

Research how attackers can reach and influence software through its dependencies, code, and exposed features.

- Study how ML, LLMs, and agents can support static application security testing (SAST), software composition analysis (SCA), infrastructure-as-code (IaC) analysis, and software bill of materials (SBOM) enrichment.
- Investigate whether attack patterns, code reachability, and execution context can reduce false alerts from isolated rule matches and help reviewers focus on exploitable issues.
- Research deterministic attack-path discovery across code, dependencies, configuration, and permissions. Record the conditions each path requires and the evidence needed to validate it.
- Map inputs from contact and lead forms, issue reports, support tickets, uploads, APIs, webhooks, chatbots, and agents. Trace how those inputs can reach internal data or actions.
- Study malicious and vulnerable packages, transitive dependencies, installation scripts, dependency hijacking, typosquatting, protestware, and maintainer history.
- Assess how SBOMs, attestations, Sigstore, in-toto, and SLSA can help verify where software came from and how it was built.

- Explore persistent memory and retrieval using retrieval-augmented generation (RAG) and knowledge graphs, with batch and streaming pipelines for data and context used by LLMs and agents.
- Study graph learning, sequence models, and anomaly detection for dependency exploitability, CI/CD drift, and unusual log behavior.
- Research developer copilots that explain findings, suggest secure fixes and pull requests, generate policies as code, and explain misconfigurations. Examine MCP integrations and VibeCoding workflows.
- Define reference test sets and evaluation measures: precision and recall at fixed alert budgets, attack-path coverage, developer acceptance, time to signal, and cost per scan. Use repeatable tests and feedback to check detection quality.

Example scenarios include Log4j exploitation, Shai-Hulud package compromise, malicious issue titles, poisoned CRM submissions, and instructions hidden in support tickets.

SARAD

Applied Machine Learning Research

- Co-developed SARAD to detect anomalies in synthetic aperture radar images using RX statistics, autoencoder reconstruction, and combined anomaly scores.
- Helped define the research problem, run experiments, and analyze anomalous images and model failures.
- Measured detection performance using ROC-AUC and precision-recall metrics.
- Improved GPU data loading using window batching and TFRecords, and tuned the training configuration.

PROFESSIONAL EXPERIENCE

Technology company

Staff AI Engineer · Contract engagement | End of 2025-Present

Promoted from Senior AIOps Engineer in 2026.

- Lead the technical design and delivery of AI automation and agent workflows.
- Build and deploy ML models, LLM applications, and agents for incident investigation, early warning, and controlled automation.
- Built incident-response agents that combine alerts, Prometheus metrics, logs, and runbooks for investigation and controlled remediation. Connected the workflow to issue tracking and team notifications.
- Developed early-warning detection using seasonal baselines, robust statistics, MAD, EWMA, CUSUM, and burn-rate methods.
- Tested detection against past incidents and operational feedback. Reduced repeated and low-value alerts using deduplication, severity thresholds, recurrence checks, and cooldowns.
- Designed shared AI services with Encore, an AI gateway, and PostgreSQL; model routing through LiteLLM; tracing in Langfuse; and workflows in Temporal.
- Expanded signal discovery and improved the reliability and monitoring of AI services and detection pipelines.

Check Point

Senior DevOps and MLOps Engineer | 2024-End of 2025

- Automated model training, validation, and deployment using SageMaker, Kubeflow, and MLflow.
- Built repeatable cloud and Kubernetes environments using Terraform and Ansible.

- Set up monitoring for models, pipelines, and infrastructure using Prometheus, Datadog, and Grafana.
- Worked with data scientists and engineers on model integration, scaling, and reliable production operation.
- Developed CI/CD pipelines and infrastructure automation for repeatable software releases.

Ezbob

Senior DevOps and MLOps Engineer | 2022-2024

- Built infrastructure to run ML algorithms over large datasets.
- Optimized and parallelized ML algorithms on a compute grid to meet latency requirements.
- Added ML capabilities to deployment automation and operational workflows.

Pagaya

Principal DevOps Software Engineer | 2021-2022

- Built release and deployment tools for production applications and ML decision systems.
- Developed ETL pipelines using Apache Airflow and AWS Glue.
- Worked on capacity planning, high availability, performance tuning, and monitoring.
- Built alerts and monitoring to track system behavior and operational performance.

Pyramid Analytics

DevOps, DataOps and Big Data Automation | 2020-2021

- Supported ML model deployment, data acquisition, and processing of large datasets.
- Developed visualization tools for data exploration and model verification.

Infibond

DevOps and DataOps Engineer | 2019-2020

- Built infrastructure supporting production ML deployment and model verification.
- Worked on data validation, access controls, security, and backup processes.

Colabo

Senior Automation and DevOps Engineer | 2018-2019

- Developed automated testing, integration and regression frameworks, and engineering workflows.

Ping Identity

Automation and Infrastructure Engineer | 2016-2018

- Developed automated processes and integration testing using Java, Python, Ruby, Appium, and Selenium.

TECHNICAL SKILLS

Data science and machine learning

Statistical analysis, predictive modeling, anomaly detection, time-series methods, feature engineering, model development, experimental evaluation, computer vision, precision-recall analysis, and failure analysis.

Languages and ML frameworks

Python, TypeScript, SQL, Bash, PyTorch, TensorFlow, scikit-learn, pandas, and NumPy.

LLMs and agentic systems

Vertex AI, Bedrock, Claude, Gemini, OpenAI, LiteLLM, Langfuse, LangChain, LangGraph, CrewAI, Temporal, agent orchestration, tool integration, and evaluation.

MLOps and data engineering

SageMaker, Kubeflow, MLflow, Anyscale / Ray, Airflow, AWS Glue, ETL pipelines, PostgreSQL, Redis, Neo4j, and BigQuery.

Infrastructure and observability

AWS, GCP, Kubernetes, Docker, Terraform, Ansible, CI/CD, Prometheus, Grafana, and Datadog.

Security research

Application exposure, code reachability, attack-path analysis, SAST, SCA, IaC security, SBOMs, dependency trust, prompt injection, MCP security, guardrail evaluation, YARA, and Opengrep.

EDUCATION AND DOCTORAL RESEARCH

PhD Researcher in Computer Science

University of Greenwich - Registered researcher | October 2026-Present

Doctoral research: Chagu and agentic AI security · ATAS UK clearance.

Research in AI security, provenance, and accountability.

Specialist Diploma in Economics

Stavropol State Agrarian University | 2004

Relevant studies: mathematics, statistics, and quantitative methods.

MBA

Università di Bologna | 2013

CERTIFICATIONS AND FELLOWSHIP

- Advanced Data Science - Y-Data
- Big Data Engineering - NAYA College
- AI Performance Engineering Fellowship - Nebius Academy
- Nebius Agentic AI Builder - Nebius Academy
- AWS Certified Solutions Architect - AWS
- Machine Learning Certificate - AlgoExpert

TECHNICAL WRITING AND CONTRIBUTIONS

- Spoke at Neo4j NODES 2025 on graphs, AI, and security.
- Write about AI security, anomaly detection, observability, and engineering.
- Publish research prototypes and ML code through Chagu, SARAD, and related projects.
- Contributed to Nebius and JetBrains Academy